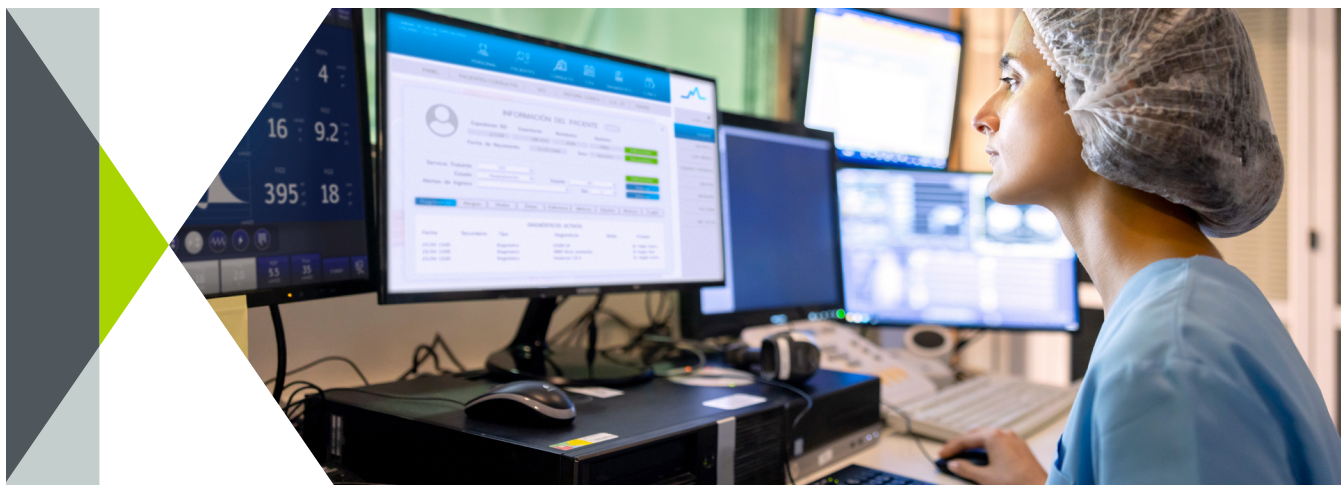




EXECUTIVE INSIGHTS

Compliance in Pharma: Navigating Increased Complexity and Strategic Opportunities

A market framework for investors and operators

From back office to boardroom

Pharmaceutical compliance was not always a boardroom concern. For decades, it operated as a labor-intensive, regionally managed function: paper-based records, reactive audit preparation and reporting through operational rather than executive channels. Three forces changed that: **Regulatory exposure escalated** (making compliance failures existential risks), **globalization** created divergent requirements across dozens of jurisdictions and **digitization** turned every interaction channel into a compliance surface.

The result was the emergence of the chief compliance officer (CCO) as a C-suite role, a development of only the last roughly 15 years. What had been a local, diffuse cottage-industry function now needed a single owner; the CCO's mandate is to hold one globally accountable source of truth on compliance, yet that source of truth has to connect disparate functions (clinical, manufacturing, commercial, safety), each shaped by different, more complicated and more varied local and regional requirements. Signing off on one coherent compliance posture built from dozens of systems, regions and asset types is precisely what makes the role so hard, and this presents a market opportunity.

Compounding complexity

These forces continue to intensify. The Food and Drug Administration's January 2025 guidance on scientific information on unapproved uses and an approximate 40% year-over-year increase in enforcement actions from the Office of Prescription Drug Promotion signal an aggressive regulatory posture. COVID-19-era digital engagement, such as remote sampling, virtual events and compliant chat, is now permanent, but each channel introduces distinct compliance risks (e.g., with the General Data Protection Regulation, the Prescription Drug Marketing Act and digital monitoring). And global consistency with local specificity remains fiendishly difficult; the complexity is not additive but multiplicative.

Mapping the compliance technology landscape

To understand the competitive dynamics of pharma compliance technology, L.E.K. Consulting maps the landscape along two dimensions: **the pharmaceutical value chain** (from research and translational realms through clinical, manufacturing, commercial and safety/pharmacovigilance (PV) areas) and **tech stack depth** (from enterprise-level systems down through functional platforms to deep workflow and machine-level tools).

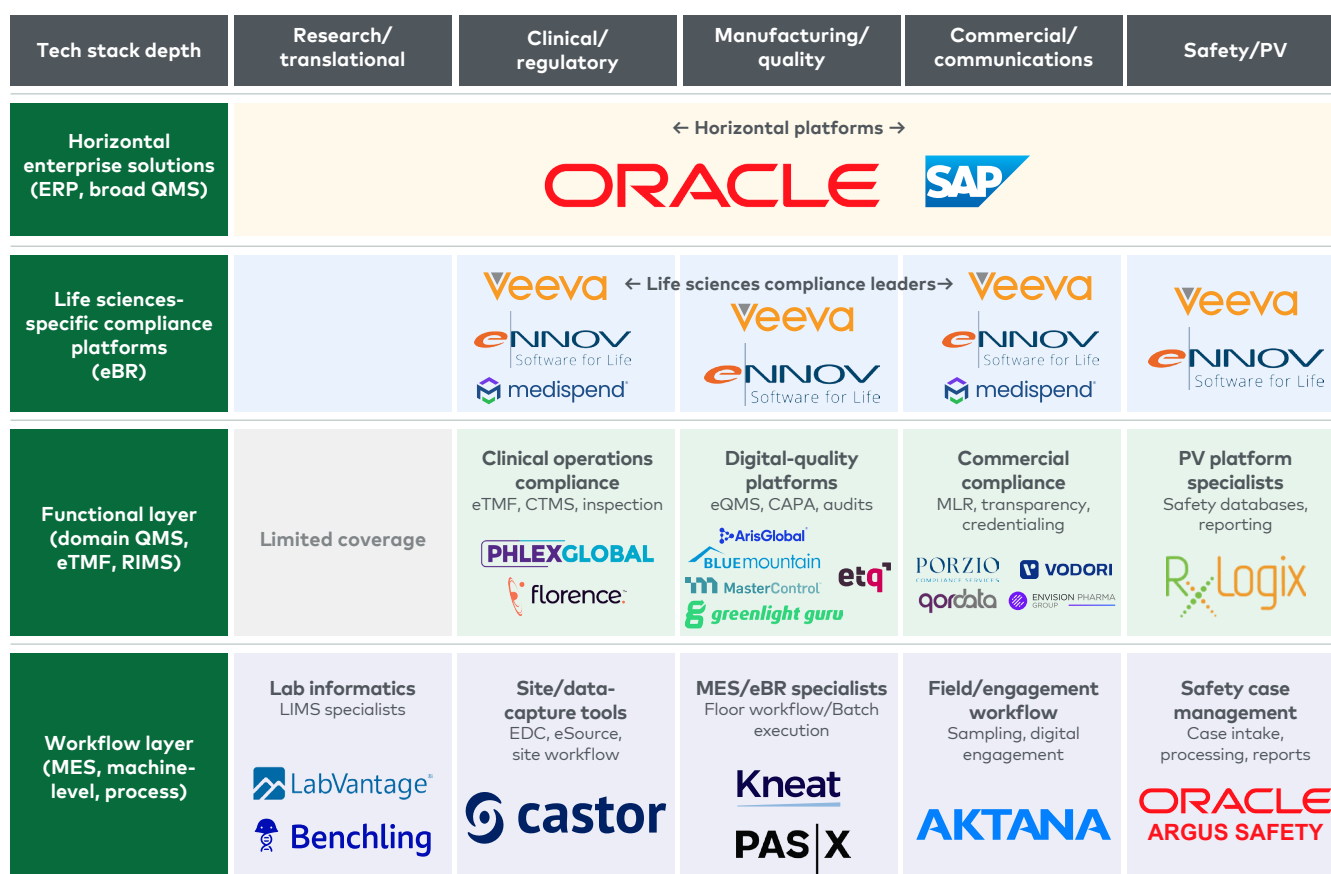
A critical pattern holds across this landscape: The higher in the stack, the more that horizontal and enterprise-spanning players compete effectively. The deeper you go, the more workflow-specific the requirements — and the more onerous the compliance burden — become. This is precisely why horizontal enterprise players have historically struggled to penetrate deep life sciences compliance. The same workflow specificity that keeps horizontal platforms out is precisely what creates durable, defensible niches for the specialist providers that own that depth. For example, FedEx and UPS spent years trying to break into clinical trial logistics and largely failed because the workflow specificity was fundamentally different from that of general logistics. The same dynamic applies throughout pharma compliance.

This is also why a single unified system is not the answer at every layer. Every customer wants one source of truth, and that ambition holds at the top of the stack, where enterprisewide oversight is the point. But the deeper the workflow, the more configuration and process specificity outweigh the appeal of standardization. Competition has fragmented not because vendors chose different markets but because end customers define compliance differently depending on where they sit in the stack, and providers that win at one layer compete on a different basis than those that win at another.

Figure 1

The compliance technology landscape

Mapping player types across pharma domains and tech stack depth*



*Representative players shown for illustrative purposes; not comprehensive

Note: PV=pharmacovigilance; ERP=enterprise resource planning; QMS=quality management system; eBR=electronic batch record; eTMF=electronic trial master file; RIMS=regulatory information management system; CTMS=clinical trial management system; eQMS=electronic quality management system; CAPA=corrective and preventive action; MLR=medical, legal and regulatory; MES=manufacturing execution system; LIMS=laboratory information management system; EDC=electronic data capture

Source: L.E.K. research and analysis

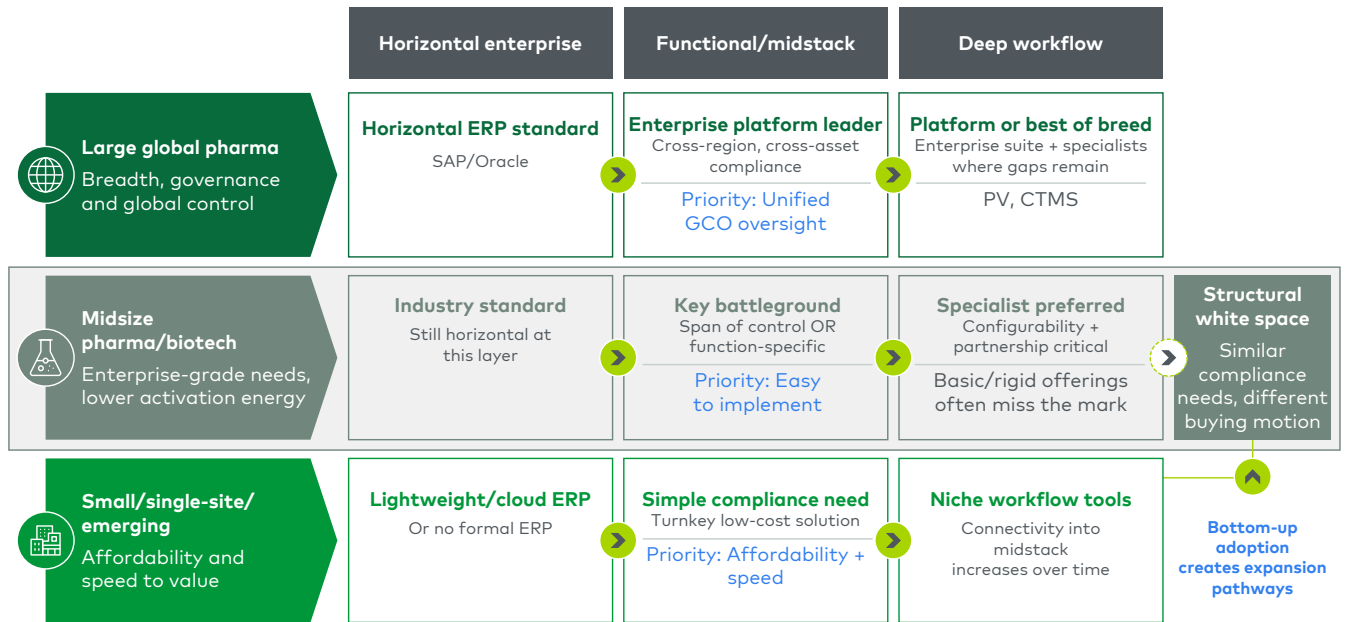
The landscape reveals several structural realities. Enterprise platform leaders (most notably Veeva) stretch across domains at the platform layer, providing the unified cross-function, cross-region compliance that CCOs demand, a position distinct from those of horizontal enterprise solutions such as SAP and Oracle, which lead the underlying enterprise resource planning layer but lack the workflow depth to penetrate below it. Functional-layer specialists — companies such as MasterControl and ETQ as well as domain-specific PV and clinical trial management system providers — compete in the middle band, offering deep expertise in specific compliance functions. At the deepest workflow layer, a long tail of niche players serve highly specific process and machine-level compliance needs. The market is also starting to see aggregators, such as MediSpend and RLDatix, bring together multiple systems and sell them independently or as middleware rather than as a single unified platform in the manner of Veeva.

Customer size and buying criteria

The landscape map explains **what** competes where. But understanding **who buys what** requires a second lens: customer size. Large global pharma, midsize biotech and small or single-site companies have fundamentally different buying criteria at each layer of the stack, and these differences create distinct market opportunities.

Figure 2

Customer size x buying criteria



Note: ERP=enterprise resource planning; GCO=global compliance officer; PV=pharmacovigilance; CTMS=clinical trial management system
Source: L.E.K. research and analysis

The critical insight for investors and operators is this: Even if you cede the large-enterprise market to established platform leaders, there is a substantial and underserved mid-market. These companies often have similar span-of-control and functional compliance needs as their larger counterparts, but they require **lower activation energy** — faster implementation, more configurable solutions, genuine partnership and competitive pricing. The “take it or leave it” approach of enterprise platform leaders’ simplified offerings (e.g., Veeva Basic and QuickVault) does not meet mid-market needs. This structural gap is where significant value creation potential exists.

At the smallest end, emerging pharma and single-site operators increasingly adopt specialist workflow tools that, over time, build connectivity upward into the functional midstack. This bottom-up adoption pattern creates organic expansion opportunities for specialist providers as well as potential roll-up targets for investors building mid-market platforms.

Three winning strategies

The intersection of the landscape map and customer size dynamics produces three distinct paths to competitive advantage in pharma compliance technology (see Figure 3).

Figure 3

Three winning strategies in compliance technology

Strategy 1	Strategy 2	Strategy 3
Enterprise platform Win by spanning functions, regions and assets on a single platform. Serve the CCO's need for unified oversight and one-click reporting. Who wins: Large, established platform providers with built-from-the-ground-up architectures Competitive edge: Integration depth, regulatory breadth, switching costs	Easy-to-implement mid-market Win by offering lower activation energy. Configurable, partnership-oriented solutions that deliver compliance value quickly without enterprise complexity. Who wins: Mid-market platforms targeting companies priced out of or underserved by enterprise leaders Competitive edge: Speed to value, configurability, customer intimacy, competitive pricing	Specialist workflow Win by going deeper than anyone else in a specific compliance domain. Own a function (PV, manufacturing QMS, eTMF) with unmatched workflow expertise. Who wins: Vertical specialists with deep domain knowledge and loyal customer bases in specific functions Competitive edge: Workflow depth, regulatory domain expertise, increasing connectivity upward into midstack

Note: CCO=chief compliance officer; PV=pharmacovigilance; QMS=quality management system; eTMF=electronic trial master file
Source: L.E.K. research and analysis

These strategies are not mutually exclusive over time; specialist workflow players can evolve into mid-market platforms and mid-market platforms can build toward enterprise scope through M&A. At any given moment, however, a provider's competitive credibility depends on clarity about which strategy it is executing. The most common strategic error in this market is trying to be all three simultaneously.

Structural shifts reshaping the landscape

Industrialization of compliance

The most fundamental shift is the industrialization of compliance itself. Documents arrive in electronic formats. Systems are designed to be proactively audit-ready rather than retrospectively assembled when an audit notice arrives. This is a philosophical change, from being "prepared when asked" to being "always prepared," and the technology providers enabling that shift command premium positioning.

AI as competitive necessity

The scale of digital interactions makes manual oversight untenable. Artificial intelligence (AI) and natural language processing have moved from differentiators to table stakes. Veeva's April 2025 "Veeva AI" investments, Within3's Launch Intelligence analytics platform and a growing cohort of AI-native startups are all responding to the same reality. Providers that fail to embed AI meaningfully will find themselves at a structural disadvantage. AI itself is changing quickly, but regulatory compliance is a measured adopter by nature. The function exists to produce traceability, verifiable truth and named accountability, and none of those are fully delegable to a system whose reasoning cannot yet be fully audited. In practice, that makes AI most valuable today as an accelerator within existing human-owned workflows, such as automated flagging of promotional content for review or continuous generation of audit-ready documentation, rather than as a wholesale replacement for the underlying compliance architecture. Zoom out and this is one instance of a broader economic shift: As the cost of complex interpretation, analysis and judgment that once required trained specialists continues to fall, that capability moves into the software itself, a dynamic reshaping pharma services well beyond compliance alone.

From fragmentation to consolidation

The era of managing compliance through disconnected niche solutions is ending, driven by CCOs demanding unified oversight and by private equity sponsors recognizing the roll-up opportunities. MediSpend's 2025 acquisitions (of Real Life Sciences in January and Alucio in August, plus its merger with RLDatix Life Sciences in September) exemplify the strategy. Consolidation is most active in the specialist and mid-market segments, where specialist depth is being assembled into broader capabilities.

Conclusion

The pharmaceutical compliance technology market is at an inflection point. Tech stack depth and customer size define the competitive boundaries. Enterprise platform leaders dominate broad-span compliance for global pharma. The mid-market is underserved and demands lower activation energy. Specialist workflow players own the deep stack and are building connectivity upward. And AI is reshaping every layer.

For investors and operators, the three winning strategies (enterprise platform, easy-to-implement mid-market and specialist workflow) provide a clear framework for identifying defensible positions and executing against structural tailwinds. Providers that understand

which strategy they are pursuing, and execute against it with discipline, will find compelling value creation potential over the 2025-27 horizon and beyond. For more information, please [contact us](#).

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About L.E.K.'s Life Science Enablers practice:

L.E.K. Consulting's Life Science Enablers practice advises investors and operators across the pharmaceutical services value chain, including compliance technology, contract research organizations, medical communications and healthcare data and analytics. Our team brings deep domain expertise in market segmentation, competitive positioning and transaction support for life sciences services and technology companies.

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L.E.K. Consulting is a premier global strategy consulting firm that partners with clients to solve their most critical business challenges and deliver high-impact outcomes. We bring deep industry expertise and rigorous, evidence-based insight to global corporations, growth companies and private equity investors, helping them turn strategy into action. Founded more than 40 years ago, L.E.K. now operates across 27 offices worldwide. Learn more at lek.com.

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